

8 In this question you must show detailed reasoning.

You are given that i is a root of the equation $z^4 - 2z^3 + 3z^2 + az + b = 0$, where a and b are real constants.

(a) Show that $a = -2$ and $b = 2$. **[4]**

(b) Find the other roots of this equation. **[7]**